

FqRL: Financial Advisor using Deep Q Learning(DQN)

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Abstract—This paper presents FqRL, a reinforcement learning-based system for financial decision-making using Deep Q-Learning (DQN). A custom financial environment simulates income, expenses, investments, inflation, and market regimes. The agent learns optimal strategies to allocate wealth among spending, saving, and investing while balancing growth, satisfaction, and financial risk. Results show consistent asset growth and adaptive decision-making across varying market conditions.

Index Terms—Reinforcement Learning, Deep Q-Network, Financial Decision Optimization, AI Finance

I. INTRODUCTION

Traditional financial planning systems rely on static rules and lack adaptability to dynamic economic conditions. Reinforcement Learning (RL) provides a framework where agents learn optimal strategies through interaction with an environment.

In this work, we develop a financial RL agent that learns to manage income, expenses, and investments under uncertain market conditions. The model uses Deep Q-Learning (DQN) to optimize long-term rewards and adapt its decision-making strategy over time.

II. SYSTEM ARCHITECTURE

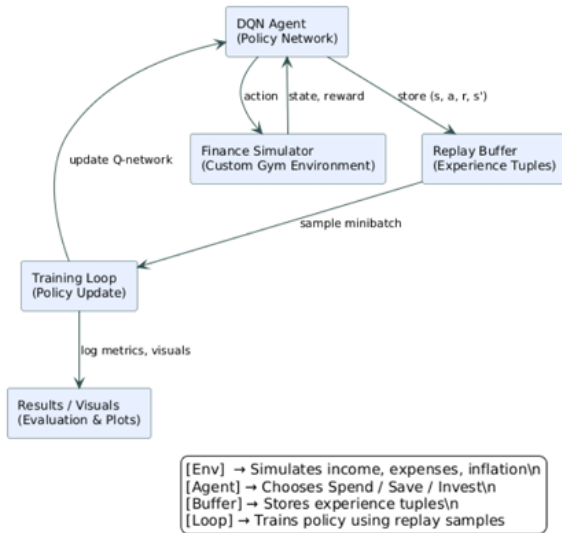


Fig. 1. System architecture of the RL-based financial advisor

The system consists of three main components:

- **Environment (Finance Simulator):** Simulates financial conditions such as income, expenses, and market behavior.
- **DQN Agent:** Learns optimal actions (spend, save, invest).
- **Replay Buffer:** Stores past experiences for stable learning.

The agent interacts with the environment, stores experiences, and updates its policy using sampled data.

III. METHODOLOGY

A. Custom Financial Environment

A Gym-compatible environment is designed where each episode represents 36 months of financial activity.

The state is defined as:

$$S_t = [balance, income, expenses, market_return, inflation, risk, goal] \quad (1)$$

The action space:

$$A_t \in \{0 : Spend, 1 : Save, 2 : Invest\} \quad (2)$$

B. Market Regime Simulation

To improve realism, the market is modeled using regimes instead of random noise.

Bull Market:

$$R_{bull} \sim \mathcal{N}(0.01, 0.02) \quad (3)$$

Bear Market:

$$R_{bear} \sim \mathcal{N}(-0.02, 0.04) \quad (4)$$

These regimes are combined to simulate structured financial trends.

C. Crash Event

A shock event is introduced:

$$R_t = -0.30 \quad (5)$$

This simulates sudden market crashes and tests model robustness.

D. Reward Function

The reward is defined as:

$$R_t = \alpha G_t + \beta S_t - \gamma F_t \quad (6)$$

where:

- G_t = portfolio growth
- S_t = user satisfaction
- F_t = financial stress

E. Learning Algorithm

The DQN updates its Q-values as:

$$Q(s, a) = Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)] \quad (7)$$

Experience replay and target networks stabilize learning.

IV. RESULTS

The model was evaluated across multiple simulated financial episodes to assess learning performance, policy behavior, and financial outcomes.

A. Training Reward Progression

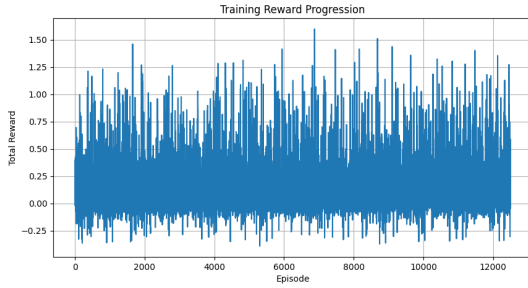


Fig. 2. Training reward progression over episodes

The training reward curve reflects the learning progress of the agent over time. Initially, the rewards fluctuate significantly due to exploration and unstable policies. As training progresses, the rewards stabilize within a bounded range, indicating convergence of the learning process.

The presence of fluctuations is expected due to stochastic components such as market returns, income variation, and inflation. Overall, the trend suggests that the agent successfully learns a consistent policy.

B. Agent Action Distribution

The action distribution illustrates how frequently each decision (spend, save, invest) is selected by the agent after training.

The results show that the agent adopts a relatively conservative strategy, prioritizing spending and saving while limiting investment exposure. This behavior reflects the balance enforced by the reward function, which penalizes excessive financial risk while encouraging stability and user satisfaction.

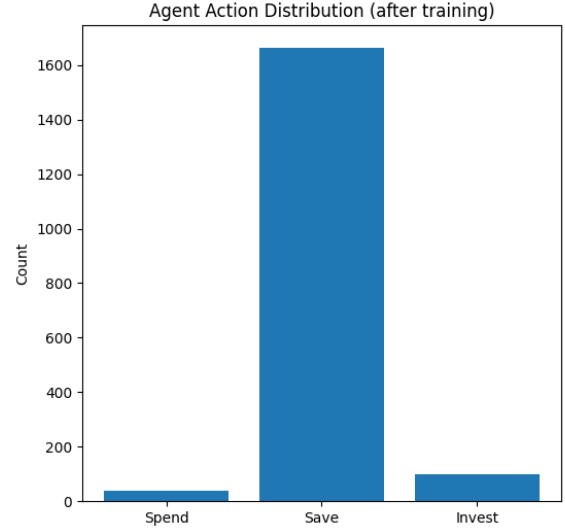


Fig. 3. Distribution of actions chosen by the trained agent

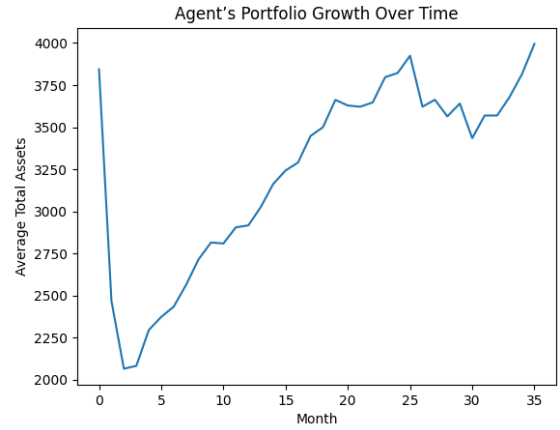


Fig. 4. Portfolio value over time

C. Portfolio Growth Curve

The portfolio curve represents the total assets accumulated over the simulation period. The upward trend indicates that the agent is able to consistently grow its wealth over time.

Small fluctuations in the curve correspond to variations in market conditions and stochastic income patterns. The absence of large sustained declines demonstrates that the agent maintains a balanced strategy and avoids excessive risk.

Overall, the results confirm that the reinforcement learning agent successfully learns an adaptive financial policy that performs well under varying conditions.

D. Multi-Episode Asset Growth

This figure shows the total asset growth of the agent across multiple independent episodes under varying market

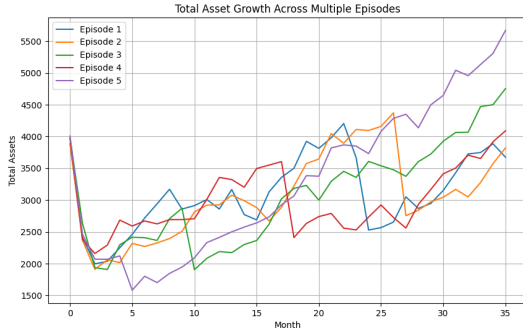


Fig. 5. Total asset growth across multiple episodes

conditions. Each curve represents a separate simulation run with different stochastic factors such as income variation, market returns, and inflation.

Despite the randomness introduced in each episode, all trajectories exhibit an overall upward trend, indicating that the learned policy generalizes well across different scenarios. The agent is able to consistently accumulate wealth while adapting to changing financial conditions.

The variation between curves highlights the uncertainty inherent in financial systems; however, the consistent growth pattern demonstrates robustness and stability of the learned strategy.

E. Action Distribution

The agent learns a balanced strategy, prioritizing spending and saving with controlled investment.

F. Portfolio Growth

The total assets increase consistently across episodes, demonstrating stable learning.

V. DISCUSSION

A. Interpretation of Results

The experimental results demonstrate that the reinforcement learning agent is capable of learning a stable and adaptive financial strategy. The agent effectively balances multiple objectives, including portfolio growth, user satisfaction, and financial risk.

From the portfolio growth curves, it is evident that the agent consistently increases total assets over time, even under stochastic market conditions. The presence of moderate fluctuations indicates that the agent is responsive to changes in income, expenses, and market returns rather than following a rigid policy.

The training reward progression further confirms that the learning process converges over time. Although the reward signal exhibits variability due to randomness in the environment, the bounded nature of the rewards indicates policy stability.

B. Behavioral Analysis of the Agent

The action distribution suggests that the agent adopts a relatively conservative financial strategy. It prioritizes spending and saving over aggressive investing, which aligns with the reward function design that penalizes excessive risk.

This behavior indicates that the agent is risk-aware and avoids large potential losses, particularly during adverse market conditions such as bear phases or crash events. By adjusting the reward weights (α , β , γ), different financial personalities can be modeled, such as aggressive investors or conservative savers.

C. Impact of Market Regimes and Crash Events

The inclusion of market regimes significantly improves the realism of the simulation. The agent learns to adapt its strategy based on the phase of the market, reducing exposure during unfavorable conditions and benefiting during positive trends.

The crash event plays a critical role in evaluating robustness. It prevents the agent from overfitting to stable environments and encourages strategies that maintain financial resilience during extreme downturns.

D. Practical Implications

The proposed system has strong potential for real-world applications. In practice:

- Financial parameters can be personalized based on user profiles
- Real-time market data can replace simulated inputs
- The model can be deployed as an intelligent financial advisory system

This demonstrates the applicability of reinforcement learning in building adaptive and personalized financial tools.

VI. CONCLUSION

This paper presented FqRL, a reinforcement learning-based framework for financial decision-making using Deep Q-Learning (DQN). The system models realistic financial environments by incorporating income, expenses, investment dynamics, inflation, and structured market regimes.

The results demonstrate that the agent is capable of learning an effective financial strategy that balances growth, user satisfaction, and financial stability. The agent consistently increases total assets across multiple episodes while maintaining robustness under varying market conditions.

A key contribution of this work is the integration of market regime simulation and crash events, which enhances realism and enables the agent to learn risk-aware strategies. This allows the model to better reflect real-world financial behavior compared to purely stochastic simulations.

Despite its effectiveness, the current system has certain limitations, including reliance on synthetic data and a simplified action space. These limitations provide opportunities for further improvement.

Future work can focus on:

- Integration of real-world financial datasets and APIs

- Use of advanced reinforcement learning algorithms such as PPO and SAC
- Personalization of reward functions based on user behavior and preferences
- Deployment as a real-time financial advisory application

In conclusion, this work highlights the potential of reinforcement learning as a powerful tool for adaptive and intelligent financial decision-making systems.

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